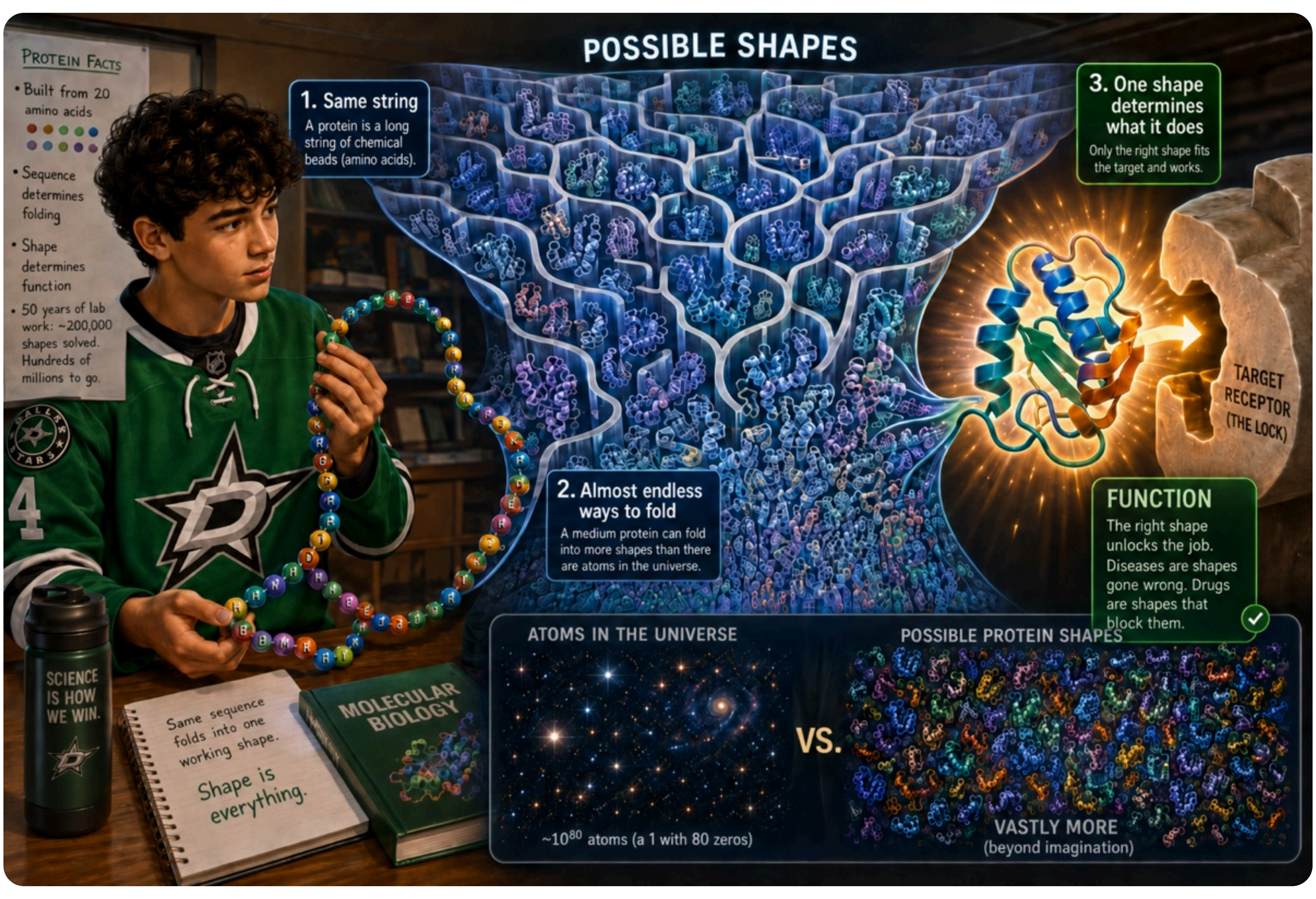
EMBRACE THE FUTURE

Big Upside

We've said it many times: AI is a giant calculator running math at a scale nobody can picture. But that calculator can do some amazing things. It solved a problem that had beaten biology for fifty years.

THE PROTEIN FOLDING PROBLEM

Your body runs on proteins. Think of them as tiny machines. They digest your food, carry oxygen, fight off infections, and build muscle. Each one starts as a long string of chemical beads, and then that string folds itself into a specific 3D shape in a fraction of a second.



That's the fifty-year problem in one picture. The shape is the whole thing: diseases are shapes gone wrong, and drugs are shapes that block them. But a medium protein can fold more ways than there are atoms in the universe, so predicting shapes was brutal. Fifty years of lab work solved about 200,000, out of hundreds of millions.

THE GUY WHO SOLVED IT

Demis Hassabis

Co-founder of DeepMind

1989	Becomes a chess master at the age of 13.
1994	At age 17 he co-designs Theme Park, a video game that sells millions of copies.
2009	Earns a PhD in neuroscience, because he wanted to understand how the brain works well enough to build one.
2010	Founds an AI company called DeepMind.
2014	Google buys DeepMind.
2020	**DeepMind solves protein folding.** The fifty-year problem, essentially finished.

DeepMind's predictions came back about as accurate as real lab experiments.

Then DeepMind did the part that actually mattered: they gave the answers away, free to everyone.

Fifty years vs one release

HUMANS
50 YEARS OF LAB WORK

~200,000
protein shapes solved

DEEPMIND
ONE RELEASE, 2022

~200,000,000
protein shapes solved, free to everyone

1,000× more shapes, released in one go

More than two million users in 190 countries have pulled from that database. In 2024, Hassabis won the Nobel Prize in Chemistry for it.

> "I've dedicated my career to advancing AI because of its unparalleled potential to improve the lives of billions of people."
>
> — DEMIS HASSABIS · On winning the 2024 Nobel Prize in Chemistry

MORE OF THE BIG UPSIDE

You learned that AI is a pattern machine. Smart people are pointing AI at problems with too many possibilities for humans to search by hand. And, the results are saving lives.

New antibiotics

MIT researchers screened thousands of compounds with AI and found one that kills a superbug almost nothing else touches. They named it abaucin.

Faster weather forecasts

One computer creates a global forecast in about a minute. The old way took hours on a supercomputer. Faster forecasts mean more people can leave dangerous areas before the bad weather hits.

Materials that don't exist

Batteries, solar panels, and computer chips all use crystals. Better crystals make them all better. Demis' company, DeepMind, predicted 380,000 crystals worth making.

Eyes and ears

Blind users point a phone and AI reads the world: the right bus, the food on the menu, the expiration date of their food. AI does the same for deaf users, turning sound into live captions. **Shipping now. FREE!**

Early flood warnings

Free warnings of upcoming floods in dozens of countries, including places where they don't have river gauges. People know days in advance, instead of minutes.

Catching what doctors miss

In a Swedish trial with more than 100,000 women, AI-supported screening detected more breast cancers, with no statistically significant increase in false positives.

Every one of those is true. So the next time someone asks, "What good does AI do for society?", you have an answer ready: **"For starters, it's saving lives."**

One more thing about Demis. His story doesn't start in a lab. It starts with a kid who loved chess and video games. The brilliance is rare. But the attitude is a choice that you can make starting today. Whatever you're good at, point it at something that helps people. You might not win the Nobel Prize. But who are we to say you won't?

The big upside is happening today.

AI is saving lives.